

Assignment: ML (3)

Dataset:

Use Pen-Digits datasets (train dataset & test dataset) with provided splits to solve questions.

Decision tree

- 1- Generate a scatterplot matrix to show the relationships between the variables and a heatmap to determine correlated attributes, then write a summary of what you noticed.
- 2- Ensure data is in the correct format for downstream processes (e.g., remove redundant information, convert categorical to numerical values, address missing values, etc.)
- 3- Fit a decision tree to the training data. Plot the tree, interpret the results, and display accuracy and Confusion Matrix.
- 4- Try different ways to improve the decision tree algorithm (e.g., use different splitting strategies, prune tree after splitting). Does pruning the tree improves the accuracy?

Bagging

(Bagging is to generate a set of bootstrap datasets, create estimators for each bootstrap dataset, and finally utilize majority voting (soft or hard) to get the final decision.)

- 1- Apply bagging strategy to classify test set samples by using SVM and Decision Tree algorithm as base estimators. Display accuracy and Confusion Matrix.
- 2- Apply Random Forest algorithm (the baseline), then fine tune this baseline. For the number of estimators, Try 5 different values within the interval of [10, 200]. Plot accuracy vs. number of estimators.

Boosting

- 1- Use GradientBoosting classifier to classify test set samples. There are 2 important hyperparameters in GradientBoosting, i.e., the number of estimators, and learning rate. First, tune number of estimators parameter by trying 4 values in the interval of [10, 200]. Then by using the tuned value for number of estimators, tune the learning rate parameter by trying 4 values within the range of [0.1, 0.9]. Display accuracy and Confusion Matrix separately for the best value of both parameters (Number of estimators and learning rate).
- 2- Build XGBoost classifier with the same parameters that you obtained in the last one. Provide accuracy and Confusion Matrix.
- 3- Comment on Bagging and Boosting approaches.

Improving with PCA and Feature Selection:

- 1- Compare the performance of the models in terms of the following criteria: precision, recall, accuracy, F-score. Identify the model that performed best and worst according to each criterion.
- 2- Choose the best model and reduce complexity and focusing on the most important features by using Principal Component Analysis (PCA) and feature selection to make it work even better.
- 3- Evaluate how well the improved model does in precision, recall, accuracy, and F1-score. This helps us see the impact of PCA and feature selection on its performance.
- 4- Compare the enhanced model's performance with its original version. With writing a comment.